Finite Sets Solutions

Solutions by: Raindrops_drop

October 11, 2026

Exercises

1.

(a) Make a list of all the injective maps

$$f : \{1, 2, 3\} \to \{1, 2, 3, 4\}.$$

Show that none is bijective. (This constitutes a *direct* proof that a set A of cardinality three does not have cardinality four).

Proof. Define f_1 such that $f_1(i) = i$, note this is not surjective since no i will map onto 4.
Define f_2 such that $f_2(i) = i + 1$, note this is not surjective since no i will map onto 1.
Define f_3 such that $f_3(1) = 1$, and $f_3(i) = i + 1$ for the other i. Note this is not surjective since no i will map onto 2.
Define f_4 such that $f_4(1) = 1$, $f_4(2) = 2$, and $f_4(3) = 4$. Note this is not surjective since no i will map onto 3.
For each f_i we can construct 5 other injective functions by cycling what element each one gets mapped onto. Note, however, that this cycling process does not change what element in the co-domain remains not covered by f_i. As such, we conclude that there exists no bijective correspondence between these two sets. □

(b) How many injective maps

$$f : \{1, \dots, 8\} \to \{1, \dots, 10\}$$

are there? (You can see why one would not wish to try to prove directly that there is no bijective correspondence between these sets).

Proof. Note that under an injective map f, the element 1 has 10 possible choices on what to get mapped onto, the element 2 then has 9 possible choices on what to get mapped onto. Following this logic, we note that there are $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3$ possible injective maps, or simply $10!/2$ which is equivalent to 1,814,400. □

2. Show that if B is not finite and $B \subseteq A$, then A is not finite.

Proof. Let $B \subseteq A$ such that B is not finite. Assume for contradiction A is finite. Then, there exists an injective function $f : A \hookrightarrow S_n$ where S_n is a section of the positive integers. Since $B \subseteq A$, note the restriction of f, $g : B \hookrightarrow S_n$ is also injective. This implies B is finite, which is a contradiction. Therefore, it must be A is not finite. □

3. Let X be the two-element set $\{0, 1\}$. Find a bijective correspondence between X^ω and a proper subset of itself.

Proof. ⋆**Section not finished**⋆ □

4. Let A be a non-empty finite simply ordered set.

(a) Show that A has a largest element. [Hint: Proceed by induction on the cardinality of A].

Proof. If A has cardinality 1, then A has only one element, and thus said element is also the largest element. Assume now for induction that a non-empty finite simply ordered set of cardinality n has a largest element. Now consider A with cardinality $n+1$. Let $a \in A$ and define $A_n = A - \{a\}$, noting A_n has cardinality n. By our induction assumption, A_n has a largest element, call it a_n. If $a_n > a$ then a_n is the largest element of A, otherwise if $a_n < a$, then since a_n is the largest element of A_n, we have $a_n \geq p$ for all $p \in A_n$, implying thus $a > p$ for all $p \in A_n$. As such, a is the largest element of A. In either case one can conclude A has a largest element. □

(b) Show that A has the order type of a section of the positive integers.

Proof. Since A is finite, there exists a bijection from A onto a section of the positive integers S_n. If the cardinality of A is one, then the single existing bijection is trivially order preserving. Assume for induction that for a non-empty finite simply order set A of cardinality n, it has the same order type as S_n, i.e., there exists an order preserving bijection between the two sets. Consider now the case A has cardinality $n+1$. Let $a \in A$ be the largest element of A (which exists by part (a)), and consider the subset $A_n \subset A$ defined as $A_n = A - \{a\}$. Note by our induction assumption there exists an order preserving bijection $g : A_n \to S_n$. Define now $f : A \to S_{n+1}$ as $f(p) = g(p)$ for $p \in A_n$ and $f(a) = n+1$. Note f is order preserving on all $p \in A_n$ and since a and $n+1$ are the largest element of A and S_n respectively, f is also order preserving with respect to them. Thus, f is an order preserving bijection between A and S_{n+1}. □

5. If $A \times B$ is finite, does it follow that A and B are finite?

Proof. It does not follow from $A \times B$ that A and B are finite. Consider the sets $A = \mathbb{R}^\omega$ and $B = \emptyset$, then $A \times B = \emptyset$. Thus, we noted an example of a finite set created by the Cartesian product of an infinite set and the empty set. □

6.

(a) Let $A = \{1, \dots, n\}$. Show there is a bijection of $\mathcal{P}(A)$ with the Cartesian product X^n, where X is the two-element set $X = \{0, 1\}$.

Proof. Define the function $f : \mathcal{P}(A) \to X^n$ by the map $S \subseteq A$ is mapped to $(x_1, x_2, \dots, x_n)$ where $x_i = 1$ if $i \in S$ and $x_i = 0$ otherwise, noting $1 \leq i \leq n$. To show that this function is injective, assume $S_1, S_2 \subseteq A$ such that f maps both subsets of A onto the same element

of X^n. Then, if $i \in S_1$, it implies $x_i = 1$, also making it that $i \in S_2$. Thus, $S_1 \subseteq S_2$. By an analogous argument we have $S_2 \subseteq S_1$, and thus $S_1 = S_2$. Hence, f is injective.
To show f is surjective, let $(x_1, \dots, x_n) \in X^n$ be an arbitrary element of X^n. Let I be the set contain all the indexes i such that $x_i = 1$. Note then $I \subseteq A$, and thus $I \in \mathcal{P}(A)$. In particular, $f(I) = (x_1, \dots, x_n)$ by definition of f. Therefore, f is also surjective, and we thus conclude f is a bijection. □

(b) Show that if A is finite, then $\mathcal{P}(A)$ is finite.

Proof. It was proved in an earlier section that if a set has n elements, its power set contains 2^n elements. Thus, if A is finite and has n elements, then $\mathcal{P}(A)$ has 2^n elements. In particular, this shows $\mathcal{P}(A)$ is finite if A is finite. □

7. If A and B are finite, show that the set of functions $f : A \to B$ is finite.

Proof. Let A and B be finite sets. Note that any function $f : A \to B$ must map every element of A to some element in B. As such, if A has cardinality n and B has cardinality m, each element of A has m possible elements it can be mapped onto. Thus, there must be m^n total distinct functions that map A to B. As such, there is a finite total amount of functions $f : A \to B$. □